

7.8-7.11

7.8 Coherent States

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

COHERENT STATE

Time evolving

$$|\alpha(t)\rangle = e^{-i\hat{H}t/\hbar} |\alpha\rangle$$

gets to

$$= e^{-i\omega t/2} |\alpha e^{-i\omega t}\rangle$$

$$\langle \alpha | \hat{a}^\dagger = \langle \alpha | \alpha^*$$

$$\langle x \rangle = \sqrt{\frac{\hbar}{2m\omega}} 2|\alpha| \cos(\omega t + \delta)$$

$$\langle p_x \rangle = -\sqrt{\frac{m\omega\hbar}{2}} 2|\alpha| \sin(\omega t + \delta)$$

$$\begin{aligned} \langle x^2 \rangle &= \frac{\hbar}{2m\omega} \left[\alpha(t)^2 + \alpha^*(t)^2 + 2|\alpha(t)|^2 + 1 \right] \end{aligned}$$

$$\begin{aligned} \langle p_x^2 \rangle &= \frac{m\omega\hbar}{2} \left[2|\alpha(t)|^2 + 1 - \alpha(t)^2 - \alpha^*(t)^2 \right] \end{aligned}$$

$$(\Delta x)^2 = \frac{\hbar}{2m\omega}$$

$$(\Delta p_x)^2 = \frac{m\omega\hbar}{2}$$

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$$\Delta x \Delta p_x = \sqrt{\frac{\hbar}{2m\omega}} \sqrt{\frac{m\omega\hbar}{2}} = \frac{\hbar}{2}$$

7.9

hermite polynomials, differential equations, and arcane magic

7.10 parity

$$\hat{\Pi} |x\rangle = |-x\rangle$$